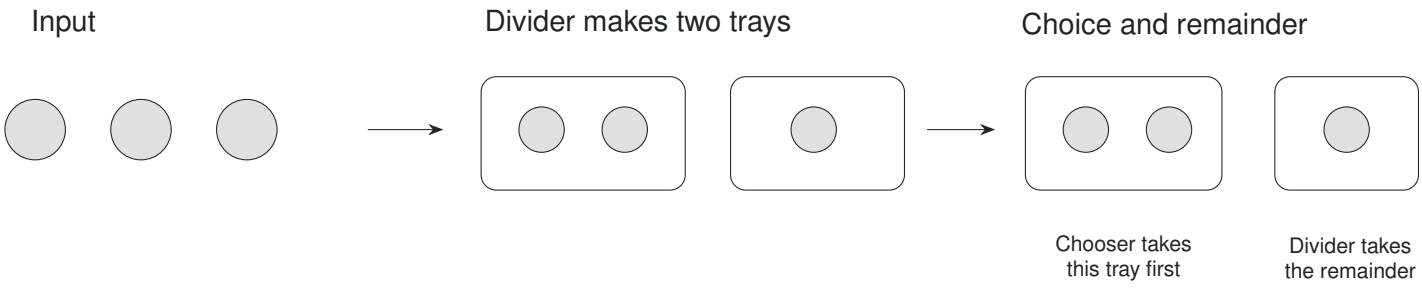
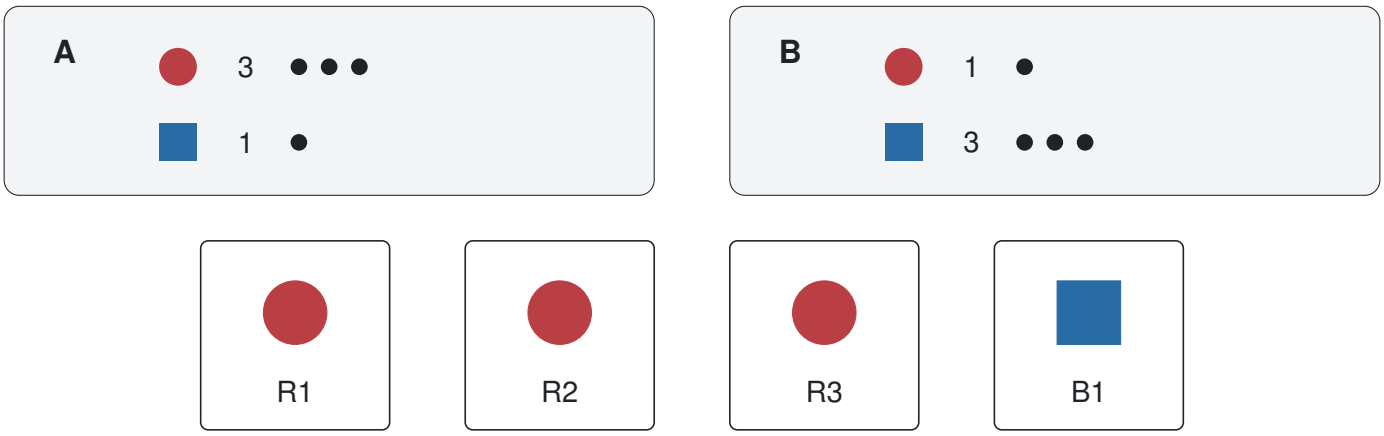
Keep each person's preference card unchanged during a trial. A tray's value is the sum of its cards' values, using the same person's preferences for every tray. Use every card once. Cards stay whole unless the problem allows cutting. Ties are allowed.




**Problem 1:** Use preference cards A and B. Divide these cards into two trays; your partner chooses a tray first, and you take the remainder. Find divisions where both people would keep their tray instead of trading. Swap who divides.




| A | B |
|---|---|

| A | B |
|---|---|

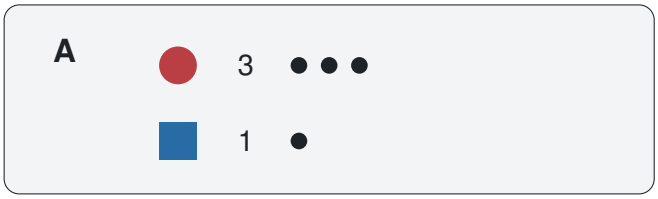
| A | B |
|---|---|

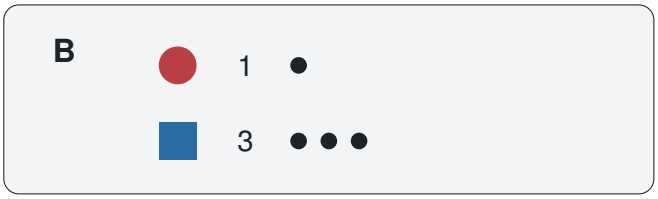
| A | B |
|---|---|

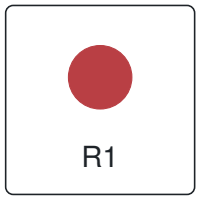
| A | B |
|---|---|

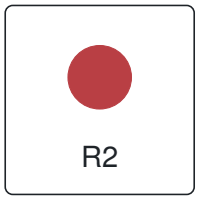
For each person, a division is *envy-free* when their own tray is worth at least as much as the other tray. It is *proportional* when their own tray is worth at least half of all the cards. Use that person's preferences for each check.

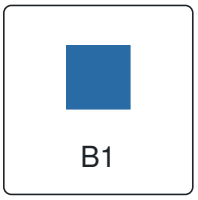
**Problem 2:** Use A and B's preference cards with R1, R2, B1 and B2. Find every way to give all four cards to A and B so that neither person envies the other. Which of these ways are proportional for both people? Different card labels count as different cards.

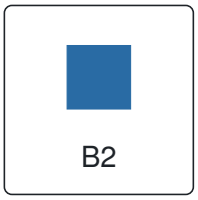



Record 1

| A | B |
|---|---|
| | |

Record 2

| A | B |
|---|---|
| | |

Record 3

| A | B |
|---|---|
| | |

Record 4

| A | B |
|---|---|
| | |

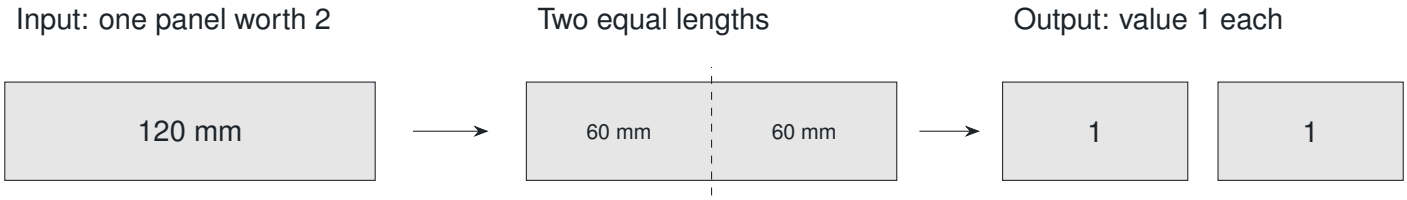
Record 5

| A | B |
|---|---|
| | |

Record 6

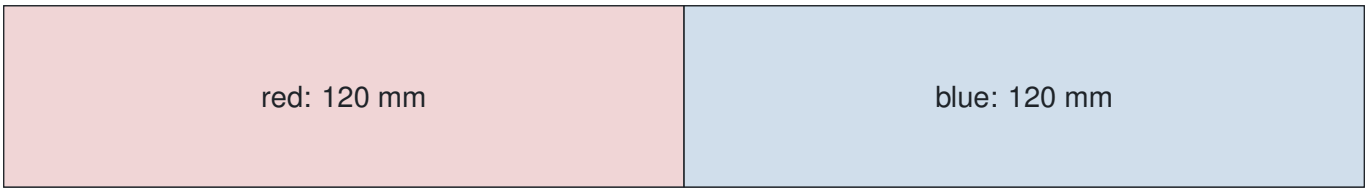
| A | B |
|---|---|
| | |

On a paper strip, the numbers on A and B's preference cards give the value of a whole 120 mm color panel. Within one color, value follows length: half a panel has half its value.



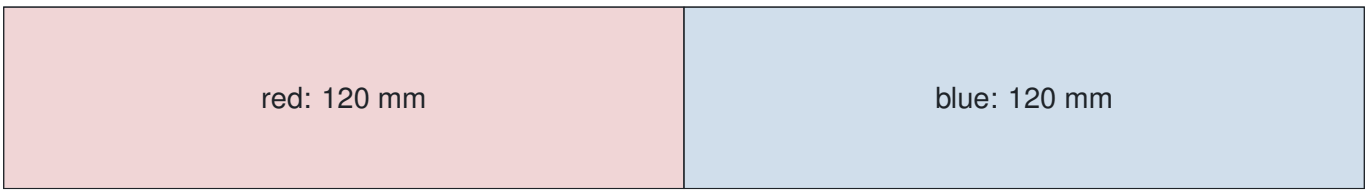

**Problem 3:** Use the full-size strips. A cuts parallel to the color join into two pieces of equal value to A. B chooses a piece first; A gets the remainder. On fresh strips, swap the roles. Mark each cut on its sketch, write its distance from the left end, and circle the chooser's piece.

A cuts




Cut: _____ mm from the left end

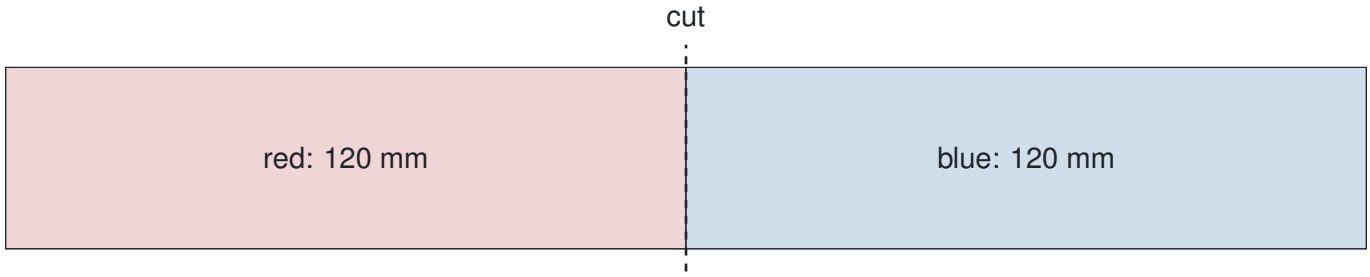
B cuts



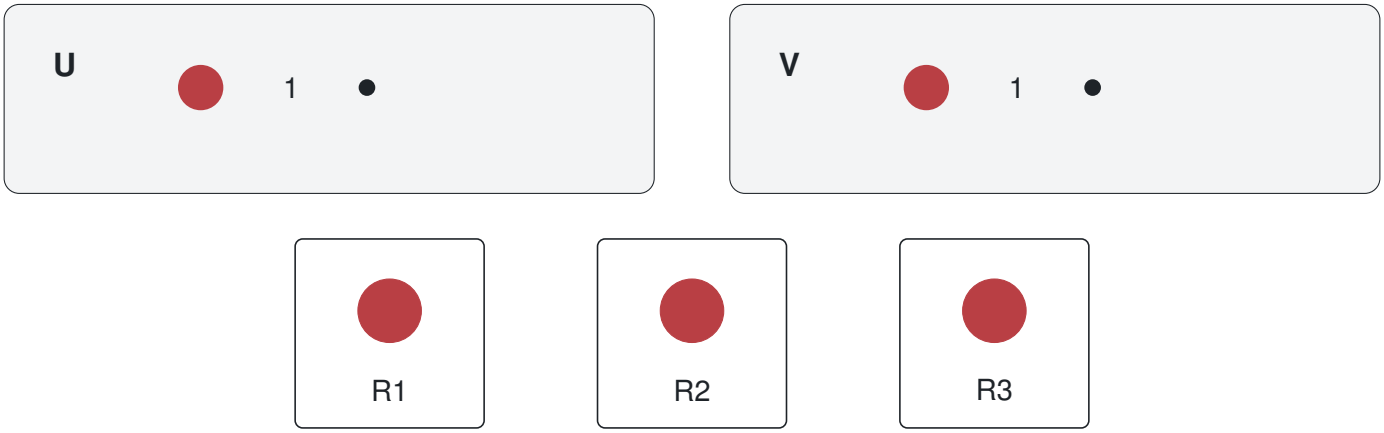

Cut: _____ mm from the left end

**Problem 4:** Two people use a splittable strip. The cutter makes pieces of exactly equal value to the cutter. The chooser takes a piece they value at least as much as the other. Must both people receive at least half of their own total value? Must neither person envy the other? Explain.

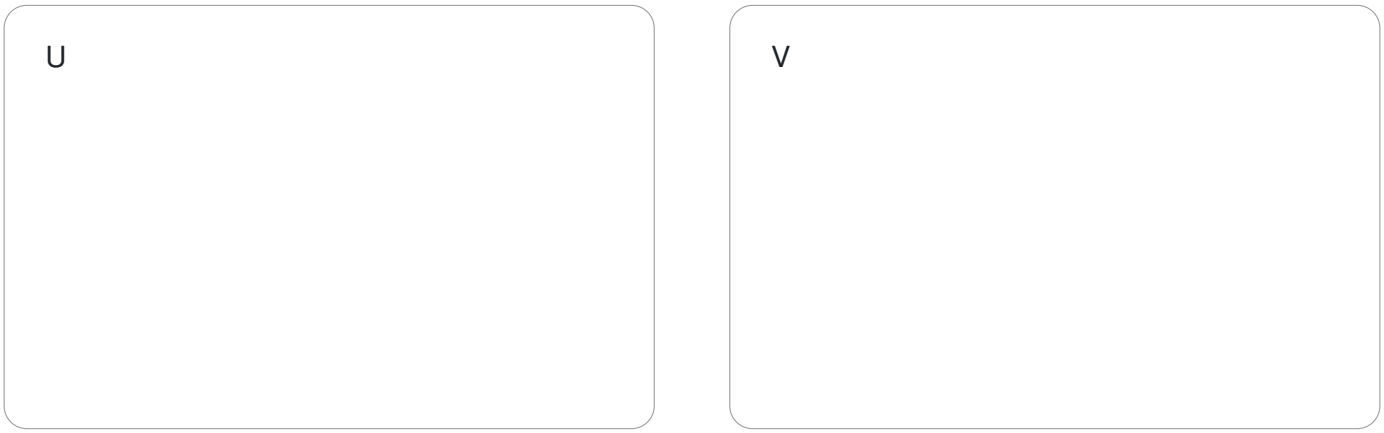
Both people now use A's strip preferences: a whole red panel is worth 3, and a whole blue panel is worth 1. The cutter cuts at the join between the colors. Does this cut-and-choose trial pass both checks?



**Problem 5:** U and V each value every red card at 1. Divide all three cards between them, keeping each card whole, so neither person envies the other. If no division works, explain why. For a second trial, replace R3 with its paper copy; you may cut that copy. R1 and R2 must stay whole. Can you now make an envy-free division?




In the second trial, R3's whole paper square is worth 1 to each person. Equal areas have equal value.



For three people, proportional means each person's own tray is worth at least one third of everything, using their own values. Envy-free means their own tray is worth at least as much as *each* other tray. Ties pass both checks.

**Problem 6:** Use the three-person preference cards below with panels X, Y and Z. A receives X, B receives Y, and C receives Z. Is this division proportional for everyone? Is it envy-free for everyone? Rearrange the whole panels to make the division envy-free.

| Observer | X | Y | Z |
|---|---|---|---|
| A | 4 | 8 | 0 |
| B | 0 | 4 | 8 |
| C | 8 | 0 | 4 |

A

X

B

Y

C

Z

Envy-free division

A

B

C

**Problem 7:** Every item goes to exactly one person, and tray values add. For two people, can a division be proportional but fail to be envy-free? Can it be envy-free but fail to be proportional? Explain your answers for any two-person division.

For three people, can a division be proportional but fail to be envy-free? Can it be envy-free but fail to be proportional? Give an example whenever the answer is yes, and explain why whenever the answer is no.